# Regime Conditional Candlestick Trading A Systematic Study of Market Structure and Pattern Effectiveness

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**Abstract**

We investigate whether conditioning candlestick pattern trading signals on a market regime detected by a Hidden Markov Model (HMM) improves risk adjusted returns relative to an unconditional baseline. A two state Gaussian Mixture HMM (GMMHMM, diagonal covariance, 3 mixture components per state) is trained on 13 financial features derived from the S&P 500 and Russell 2000 indices over 2016 to 2021 and identifies two latent market regimes, labelled *Bull* and *Bear*. Seven strategy configurations, varying long/short posture and regime filter, are backtested across three equity universes (Large Cap, Small Cap, Combined). Each is tuned independently with a two stage rolling walk forward search over 2021 to 2023 and then run once on a locked 2024 to 2025 holdout period, using a daily filtered regime label that relies only on data up to each day's close.

On the evaluation period, the regime conditioned version of a strategy generally comes out ahead of its regime agnostic counterpart on a risk adjusted basis. Across the 12 matched comparisons (four conditioned strategies, each against the agnostic strategy of the same long/short posture, on three universes), the conditioned strategy has the higher Sharpe ratio in 9, ties in 2, and trails in 1. The best strategy in every universe is the regime conditioned `LongOnly_Bull`, which trades long only on Bull days. It reaches a Sharpe ratio of 1.32 on the large cap universe against 1.11 for its agnostic baseline ($\Delta$Sharpe $= +0.21$), with a Sortino ratio of 2.10 and a maximum drawdown of 2.2%. Its Sharpe also edges SPY buy and hold (1.28) at about one ninth of SPY's drawdown, although at 5% of equity per position its absolute return is far smaller. The largest Sharpe improvements come on the short side ($+1.37$ for `LongBull_ShortBear` and $+1.33$ for `ShortOnly_Bear` on large caps), where restricting shorts to Bear days mainly cuts the losses of shorting in a rising market.

Testing the same candlestick patterns across all seven strategies also shows where the patterns themselves are strong or weak. Bullish Engulfing wins 59% of its trades in both regimes, whereas short trades opened on bearish patterns win only 37%. For the optimisation, searching all five parameters at once overfits, and fixing the three related Bollinger settings before tuning the exits cuts the gap between in sample and out of sample Sharpe from 1.43 to 0.85. Among three feature sources for regime detection (financial, macroeconomic, and combined), only the financial model produces a usable Bull and Bear split (silhouette = 0.317), while the other two collapse to a single regime. We do not claim that these strategies beat the index. Our results show that conditioning a candlestick strategy on an HMM regime improves its risk adjusted performance in most comparisons, though the gains are moderate in absolute terms and rest on a two year evaluation period.

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# 1 Introduction

Financial markets operate across qualitatively distinct states (periods of trending, mean reversion, high volatility, and low volatility), commonly referred to as market regimes. A trading strategy that performs well in a bull regime may be harmful in a bear regime. Candlestick patterns, one of the oldest forms of technical analysis, are no exception. A Hammer pattern that reliably signals a bullish reversal in a sideways market may carry little predictive power when broad sentiment is unambiguously bearish [9, 10].

Marshall, Young, and Rose [12] conduct one of the most rigorous tests of candlestick profitability and conclude that individual patterns do not generate statistically significant profits after transaction costs in the Dow Jones Industrial Average. This raises a natural question, namely *can regime conditioning rescue candlestick signals that fail unconditionally?* We study this question empirically across three equity universes and seven strategies.

Our contributions are as follows.

1. A research pipeline that joins regime detection and candlestick based trading, spanning data download, feature engineering, HMM training, and backtesting.
2. An empirical characterisation of which candlestick patterns are most effective in each regime, providing actionable guidance for practitioners.
3. A cross universe analysis comparing large cap (LMC), small cap (SMC), and combined (LMC plus SMC) performance across regimes, testing the small cap effect hypothesis in a regime conditional setting and showing that the combined universe outperforms either alone.
4. A comparison of three feature sources for regime detection (financial features alone, macroeconomic features alone, and a combination of the two), finding that the parsimonious financial model with 13 features dominates.
5. A detailed description of the pipeline with two phases (HMM training and rolling walk forward strategy optimisation) used to study regime conditional systematic trading.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the data. Section 4 details the methodology, including the two phase pipeline architecture. Section 5 describes the experimental setup. Section 6 presents backtest results. Section 7 discusses findings and limitations. Section 8 concludes.

# 2 Related Work

## 2.1 Market Regime Detection

Hidden Markov Models were introduced to financial modelling by Hamilton [1] for business cycle dating, building on the foundational HMM tutorial by Rabiner [2]. HMMs have since been applied to equity volatility regimes by Ang and Bekaert [3] and to dynamic portfolio optimisation by Nystrup et al. [4]. Guidolin and Timmermann [5] demonstrate that regime switching models with four states can capture joint equity and bond dynamics and improve asset allocation decisions. Alternative regime detection approaches include threshold autoregressive models [6], Markov switching GARCH [7], and machine learning based methods [8].

Our approach uses a parsimonious set of 13 financial features derived from the S&P 500 and Russell 2000 indices. We compare this against macroeconomic (FRED) and combined feature sources, finding that the financial only model produces the most interpretable two state regime structure.

### 2.2 Candlestick Pattern Trading

Candlestick charting originated in Japanese rice futures markets in the 18th century and was introduced to Western technical analysis by Nison [9]. Empirical evidence for candlestick predictability is mixed. Bulkowski [10] documents statistically significant predictive power for several patterns in U.S. equities. However, Duvinage et al. [11] find that many single candle patterns are not consistently profitable after transaction costs. Marshall, Young, and Rose [12] apply a rigorous bootstrap methodology to the DJIA and conclude that candlestick trading strategies do not add value over a buy and hold approach, once data snooping bias is accounted for.

We differ from prior work by conditioning pattern signals on the regime detected by the HMM, testing whether patterns that fail unconditionally may be profitable in specific market environments.

### 2.3 Regime Conditional Technical Trading

The interaction between regime and technical signal has received limited academic attention. Brock et al. [13] show that simple moving average rules have predictive power but do not condition on regime. Sullivan, Timmermann, and White [14] apply White's Reality Check [15] to a universe of technical trading rules and find that the best rules do not survive correction for data snooping. Lo, Mamaysky, and Wang [16] provide a formal statistical foundation for technical analysis by fitting kernel regressions to price data and detecting patterns algorithmically. Neely et al. [17] find that technical rules perform better during recessions, implicitly suggesting regime conditioning. Bailey and López de Prado [18] introduce the Deflated Sharpe Ratio, which adjusts for multiple testing when evaluating trading strategies. This concern is directly relevant to our search over 21 combinations. To our knowledge, ours is among the first systematic studies that explicitly maps candlestick patterns to regimes detected by an HMM across both large cap and small cap universes.

### 2.4 Small Cap Effect

Banz [19] documented that small cap stocks earn higher returns than predicted by the CAPM. Fama and French [20] formalised this as the SMB (Small Minus Big) factor. The small cap effect is known to be regime dependent [21], motivating our cross universe analysis within each regime.

## 3 Data

### 3.1 Equity Universes

We use three equity universes, fixed at a January 2022 snapshot and carried over from our earlier QuantConnect algorithm (Section 4.8). Two are the base large cap and small cap sets, and the third is their union. The lists below are exactly the tickers used in every backtest.

- **LMC (Large Market Cap).** 25 tickers from the S&P 500, spread across eight sectors so that no sector exceeds 30% of the universe. AAPL, MSFT, GOOGL, NVDA, META, AMZN, TSLA, HD, MCD, JPM, BAC, V, MA, JNJ, UNH, PFE, ABBV, CAT, HON, RTX, XOM, CVX, PG, WMT, and NEE.
- **SMC (Small Market Cap).** 15 tickers from the Russell 2000. IRMD, PCRX, IMMR, HLIT, CEVA, RAMP, SCHL, FIZZ, HEAR, PBFS, SHEN, KTOS, OSIS, MGPI, and TRUP. Five names from the original list (CPRX, CVGW, SWI, ICAD, and NATH) were later acquired or no longer have a usable price history and are left out.
- **COMBINED.** The union of LMC and SMC (40 tickers), traded from a single portfolio.

## 3.2 OHLCV Data

Daily Open, High, Low, Close, Volume (OHLCV) data for all tickers is sourced from Yahoo Finance via the `yfinance` Python library with adjusted close prices. The training period spans January 2016 to December 2021 (for HMM training), with validation and evaluation from January 2022 to December 2025.

## 3.3 Regime Feature Sources

We compare three feature sources for HMM regime detection, described below.

### 3.3.1 Financial Market Features (Primary Model)

The primary model uses 13 features derived from two major equity indices. Table 1 lists the complete feature set.

**Table 1:** Financial Feature Set (13 Features)

| Feature | Description |
|---|---|
| *S&P 500 (Large Cap), 5 features* | |
| `spx_logret_1d` | Daily log return (short term momentum) |
| `spx_logret_21d` | 21 day cumulative log return (medium term trend) |
| `spx_vol_21d` | 21 day realised volatility |
| `spx_close_zscore_21d` | Close price z score vs. 21 day mean |
| `spx_dd_63d` | 63 day rolling max drawdown |
| *Russell 2000 (Small Cap), 5 features* | |
| `rut_logret_1d` | Daily log return |
| `rut_logret_21d` | 21 day cumulative log return |
| `rut_vol_21d` | 21 day realised volatility |
| `rut_close_zscore_21d` | Close price z score vs. 21 day mean |
| `rut_dd_63d` | 63 day rolling max drawdown |
| *Cross Index Spread (SPX − RUT), 3 features* | |
| `spread_logret_1d` | Daily spread log return (size rotation) |
| `spread_logret_21d` | 21 day cumulative spread return (risk on/off tilt) |
| `spread_vol_21d` | 21 day spread volatility (size factor dispersion) |

All features use log returns for additive time aggregation properties. The feature set is designed to capture (i) short term and medium term momentum in both large cap and small cap segments, (ii) realised volatility as a proxy for risk level, (iii) mean reversion signals via z scores, (iv) tail risk depth via drawdowns, and (v) large cap vs. small cap rotation dynamics via the spread features.

### 3.3.2 Macroeconomic Features (FRED)

As an alternative feature source, we download 42 macroeconomic time series from the Federal Reserve Economic Data (FRED) API, covering interest rates (Federal Funds Rate, 2Y/10Y Treasury yields, yield curve slope), credit spreads (IG and HY), equity volatility (VIX), economic activity (unemployment, ISM Manufacturing PMI, retail sales), and money supply (M2 growth). FRED series are resampled to daily frequency via forward filling and aligned with market dates. We compute yield curve slope (10Y−2Y Treasury spread), z scores over a 252 day rolling window, and month over month and year over year changes for level series. After these transformations the 42 raw series expand to 200 engineered macro features.

### 3.3.3 Combined Feature Set and PCA

For the combined model, financial and macroeconomic features are concatenated into a single feature matrix. Principal component analysis (PCA) is applied to reduce dimensionality and decorrelate features before HMM training. Features are standardised before PCA, and we retain 50 principal components, which explain roughly 90 to 95% of the variance of the standardised feature matrix. The PCA pipeline is fitted on the training set and frozen for inference.

## 3.4 Benchmark

We use two kinds of benchmark. The passive benchmark is SPY (SPDR S&P 500 ETF), representing buy and hold investing with no strategy switching. The active benchmark is the regime agnostic version of each strategy (Table 6), which trades the same candlestick signals with the same entry and exit rules but never consults the HMM. Each regime conditioned strategy is compared with the agnostic strategy of the same long/short posture (`LongOnly_Bull` and `LongOnly_Bear` against `Agnostic_LongOnly`, `ShortOnly_Bear` against `Agnostic_ShortOnly`, and `LongBull_ShortBear` against `Agnostic_LongShort` so the difference between the two isolates the effect of conditioning on the regime.

# 4 Methodology

## 4.1 Feature Engineering

Features are computed from daily OHLCV data for SPY (S&P 500 proxy) and IWM (Russell 2000 proxy). Log returns are computed at 1 day and 21 day horizons. Realised volatility is the annualised standard deviation of daily log returns over a 21 day rolling window. The z score normalises the closing price relative to its 21 day rolling mean and standard deviation. The 63 day maximum drawdown measures the largest peak to trough decline over the trailing quarter. Spread features are the difference between SPX and RUT counterparts, capturing relative rotation between large cap and small cap segments.

All rolling statistics are computed causally (using only data available at time $t$) to prevent lookahead bias.

## 4.2 Regime Detection via Hidden Markov Model

A Hidden Markov Model posits that the observed market data is generated by a latent (hidden) discrete state, the regime, that evolves as a first order Markov chain. The model is fully specified by three objects.

1. **Initial distribution** $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$, where $\pi_k = P(s_1 = k)$.
2. **Transition matrix** $\mathbf{A} \in \mathbb{R}^{K \times K}$, where $A_{ij} = P(s_{t+1} = j \mid s_t = i)$.
3. **Emission densities** $b_k(\mathbf{x}_t) = P(\mathbf{x}_t \mid s_t = k)$, one per regime.

### 4.2.1 Emission Density (Gaussian Mixture)

We need to describe what the market data looks like inside each regime. A single Gaussian per regime would force every Bear day to resemble one average Bear day, yet real bear markets contain mild pullbacks, sharp selloffs, and quiet stabilisation days. We therefore let each regime $k$ emit the feature vector $\mathbf{x}_t \in \mathbb{R}^D$ from a mixture of $M$ Gaussians (a GMM HMM), so one regime can hold several submodes of market behaviour.

$$b_k(\mathbf{x}_t) = P(\mathbf{x}_t \mid s_t = k) = \sum_{m=1}^{M} c_{km}\, \mathcal{N}(\mathbf{x}_t;\, \boldsymbol{\mu}_{km},\, \boldsymbol{\Sigma}_{km})\,, \qquad \sum_{m=1}^{M} c_{km} = 1, \tag{1}$$

where $c_{km}$ is the weight of mixture component $m$ inside regime $k$, $\boldsymbol{\mu}_{km}$ is its mean feature vector, and $\boldsymbol{\Sigma}_{km}$ is its diagonal covariance (one variance per feature). The joint probability of a full sequence (a

hidden regime path together with the observed data) multiplies together the start probability, one transition probability per day, and one emission probability per day.

$$P(\mathbf{x}_{1:T},\, s_{1:T}) = \pi_{s_1}\, b_{s_1}(\mathbf{x}_1) \prod_{t=2}^{T} A_{s_{t-1},\, s_t}\, b_{s_t}(\mathbf{x}_t). \tag{2}$$

**Our estimated financial model.** For the financial source, model selection (Section 4.2.4) settles on $K = 2$ regimes with $M = 3$ mixture components per state over the $D = 13$ features. The estimated initial distribution and transition matrix are as follows.

$$\boldsymbol{\pi} = \begin{pmatrix} 0.00 \\ 1.00 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} A_{B\to B} & A_{B\to R} \\ A_{R\to B} & A_{R\to R} \end{pmatrix} = \begin{pmatrix} 0.980 & 0.020 \\ 0.054 & 0.946 \end{pmatrix}, \tag{3}$$

where $B$ = Bull and $R$ = Bear. The initial distribution simply records how the algorithm starts. Baum-Welch sets $\hat{\pi}_k = \gamma_1(k)$, the posterior for the first training day, which here decoded as Bear with near certainty. Both regimes are highly persistent. A Bull day has a 98.0% chance of remaining Bull, and a Bear day has a 94.6% chance of remaining Bear. This implies expected regime durations of $1/(1-0.980) = 50$ trading days (Bull) and $1/(1-0.946) \approx 19$ trading days (Bear).

Table 2 reports the weighted emission means for each regime (averaged across the three mixture components), showing how the two regimes differ in their market characteristics.

**Table 2:** Emission Means by Regime (Weighted Average Across Mixture Components)

| Feature | Bull (Regime 0) | Bear (Regime 1) |
|---|---|---|
| spx_logret_1d | +0.0008 | −0.0012 |
| spx_logret_21d | +0.019 | −0.028 |
| spx_vol_21d | 0.11 | 0.22 |
| spx_close_zscore_21d | +0.35 | −0.52 |
| spx_dd_63d | −0.03 | −0.11 |
| rut_logret_1d | +0.0006 | −0.0014 |
| rut_logret_21d | +0.015 | −0.032 |
| rut_vol_21d | 0.14 | 0.27 |
| rut_close_zscore_21d | +0.28 | −0.48 |
| rut_dd_63d | −0.04 | −0.14 |
| spread_logret_1d | +0.0002 | +0.0002 |
| spread_logret_21d | +0.004 | +0.004 |
| spread_vol_21d | 0.07 | 0.10 |

The Bull regime is characterised by positive returns, low realised volatility ($\sim$11% annualised for SPX), positive z scores (prices above their 21 day mean), and shallow drawdowns ($-3\%$). The Bear regime shows negative returns, doubled volatility ($\sim$22%), negative z scores, and drawdowns roughly four times deeper ($-11\%$). The spread features show little difference between regimes, indicating that large cap vs. small cap rotation is not a primary driver of regime assignment.

### 4.2.2 Estimation with the Baum-Welch (EM) Algorithm

All parameters $\boldsymbol{\theta} = \{\boldsymbol{\pi}, \mathbf{A}, \{c_{km}, \boldsymbol{\mu}_{km}, \boldsymbol{\Sigma}_{km}\}\}$ are estimated by the Baum-Welch algorithm [2, 22], an instance of Expectation Maximisation (EM).

**E step (Forward Backward).** Define the forward variable $\alpha_t(k) = P(\mathbf{x}_{1:t}, s_t = k)$ and backward variable $\beta_t(k) = P(\mathbf{x}_{t+1:T} \mid s_t = k)$. These satisfy the recursions below.

$$\alpha_1(k) = \pi_k \, b_k(\mathbf{x}_1), \qquad \alpha_t(k) = b_k(\mathbf{x}_t) \sum_j \alpha_{t-1}(j) \, A_{jk}, \tag{4}$$

$$\beta_T(k) = 1, \qquad \beta_t(k) = \sum_j A_{kj} \, b_j(\mathbf{x}_{t+1}) \, \beta_{t+1}(j). \tag{5}$$

The posterior state probability and pairwise posterior are then given by

$$\gamma_t(k) = P(s_t = k \mid \mathbf{x}_{1:T}) = \frac{\alpha_t(k) \, \beta_t(k)}{\sum_j \alpha_t(j) \, \beta_t(j)}, \tag{6}$$

$$\xi_t(i,j) = P(s_t = i, \, s_{t+1} = j \mid \mathbf{x}_{1:T}) = \frac{\alpha_t(i) \, A_{ij} \, b_j(\mathbf{x}_{t+1}) \, \beta_{t+1}(j)}{\sum_p \sum_q \alpha_t(p) \, A_{pq} \, b_q(\mathbf{x}_{t+1}) \, \beta_{t+1}(q)}. \tag{7}$$

**M step.** Given $\gamma$ and $\xi$, the parameters are reestimated in closed form as

$$\hat{\pi}_k = \gamma_1(k), \qquad \hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \qquad \hat{\boldsymbol{\mu}}_{km} = \frac{\sum_t \gamma_t^{(km)} \mathbf{x}_t}{\sum_t \gamma_t^{(km)}}, \tag{8}$$

where $\gamma_t^{(km)}$ is the posterior responsibility of mixture component $m$ in state $k$ at time $t$. Covariances and mixture weights are updated analogously. The algorithm iterates E and M steps until log likelihood convergence (tolerance $10^{-4}$).

### 4.2.3 Inference with the Viterbi Algorithm

At inference time, we decode the most probable regime sequence via the Viterbi algorithm [2, 23], a dynamic programming procedure. We define

$$\delta_1(k) = \pi_k \, b_k(\mathbf{x}_1), \qquad \delta_t(k) = b_k(\mathbf{x}_t) \, \max_j \, \delta_{t-1}(j) \, A_{jk}, \tag{9}$$

$$\psi_t(k) = \arg\max_j \, \delta_{t-1}(j) \, A_{jk}. \tag{10}$$

The optimal final state is $\hat{s}_T = \arg\max_k \delta_T(k)$, and the full path is recovered by backtracking, with $\hat{s}_t = \psi_{t+1}(\hat{s}_{t+1})$ for $t = T-1, \ldots, 1$.

**Viterbi versus filtered labels.** Viterbi decodes a whole sequence at once, so the label it assigns to day $t$ also depends on the observations after $t$. We use it only where hindsight is acceptable, namely Phase 1 model selection, where the silhouette scores of Section 4.2.4 are computed on the Viterbi decoded training labels. For trading we instead use the filtered label, the most probable regime on day $t$ given only $\mathbf{x}_{1:t}$. This is exactly what a live system would know at the close of day $t$, so it is the more realistic choice for real world use and keeps the backtests free of look ahead (Section 4.7.1).

### 4.2.4 Model Selection

We search over $K \in \{2, 3, 4, 5, 6\}$ states and five HMM families, namely `GaussianHMM_diag`, `GaussianHMM_spherical`, `GMMHMM_spherical_mix2`, `GMMHMM_diag_mix2`, and `GMMHMM_diag_mix3`. Each (family, $K$) pair is fitted 10 times with different random initialisations, and the best likelihood run is retained, yielding $5 \times 5 \times 10 = 250$ total fits.

Selection proceeds in two stages.

1. **Within family $K$ selection (silhouette [25]).** For each family, we select the $K$ that maximises the silhouette score on the Viterbi decoded labels in the 13 dimensional feature space. This rewards geometrically well separated clusters.
2. **Across family selection (BIC [24]).** Among the five family winners, we select the model with the lowest Bayesian Information Criterion, which penalises model complexity, defined as $\text{BIC} = -2\ln\hat{L} + p\ln T$, where $p$ is the number of free parameters.

The winner is `GMMHMM_diag_mix3` with $K = 2$ (silhouette $= 0.317$, BIC $= 31{,}946$). The next best family (`GMMHMM_diag_mix2`, $K = 2$) has BIC $= 34{,}418$, a gap of $+2{,}472$. Table 3 reports the family winners.

**Table 3:** Model Selection Family Winners (Stage 2)

| Family | $K$ | Silhouette | BIC | $\Delta$BIC |
|---|---|---|---|---|
| GMMHMM_diag_mix3 | 2 | 0.317 | 31,946 | n/a |
| GMMHMM_diag_mix2 | 2 | 0.235 | 34,418 | +2,472 |
| GMMHMM_spherical_mix2 | 2 | 0.258 | 39,809 | +7,863 |
| GaussianHMM_diag | 2 | 0.271 | 41,276 | +9,330 |
| GaussianHMM_spherical | 2 | 0.303 | 45,071 | +13,125 |

All five families independently select $K = 2$ as optimal, providing strong evidence for a two state regime structure in the financial feature space.

### 4.3 Regime Labelling

The HMM assigns numerical labels (0 and 1) to regimes, but these carry no inherent economic meaning. We assign the labels *Bull* and *Bear after the fact*, based on the emission statistics in Table 2. The regime with positive mean SPX returns and low volatility is labelled Bull, and the other is labelled Bear. This labelling is purely a convenience for discussion and does not affect the model or trading logic.

**Table 4:** Two Regime HMM Characterisation (Financial Model)

| Regime | Label | Characterisation | Occupancy (%) |
|---|---|---|---|
| 0 | Bull | Positive returns, low volatility, shallow drawdowns | $\sim$55% |
| 1 | Bear | Negative returns, high volatility, deep drawdowns | $\sim$45% |

### 4.4 Feature Source Comparison

To assess the value of the financial model with 13 features relative to alternatives, we train identical GMMHMM_diag_mix3 models on three feature sources. The results differ sharply.

**Table 5:** Regime Occupancy by Feature Source (Validation Period, 2022 to 2025)

| Feature Source | Bull (%) | Bull (days) | Bear (%) | Silhouette |
|---|---|---|---|---|
| Financial (13 features) | 55.1 | 553 | 44.9 (450d) | 0.317 |
| FRED Macro (42 series) | 100.0 | 1,003 | 0.0 (0d) | n/a |
| Combined (fin. + macro, PCA applied) | 99.9 | 1,002 | 0.1 (1d) | n/a |

The financial only model produces a meaningful Bull/Bear split (55%/45%) with the highest silhouette score (0.317). In stark contrast, the FRED model collapses to a single all Bull regime. Over the entire

2022 to 2025 validation period, which includes the 2022 bear market, the macro model assigns *zero* days to its Bear state. This occurs because macroeconomic series (unemployment, GDP, PMI) are published at monthly or quarterly frequency and forward filled to daily, creating a feature matrix that is nearly constant day to day. The HMM, trained on such smooth inputs, cannot distinguish intramonth market dynamics.

The combined model (financial + macro with PCA applied) inherits this pathology. The PCA components are dominated by the 42 low variance macro features, washing out the 13 high variance financial features. The result is a model that assigns 99.9% of days to a single regime, which makes it functionally identical to the FRED model.

We therefore adopt the financial only model as the primary regime detector. The FRED and combined models are retained as comparisons to demonstrate that "more data" does not automatically improve regime detection.

Figure 1 shows why. Projected onto its first two principal components, the macro feature space separates the historical recessions cleanly from normal conditions, yet the 2022 validation days fall entirely inside the normal region. This two dimensional projection is for visualisation only. The first two components carry about 30% of the variance of the standardised macro features, whereas the combined HMM is trained on 50 components covering roughly 90 to 95%. The picture makes one point. Even after every feature is standardised to unit variance, the macro block contributes far more features than the financial block, so the leading components are shaped by the macro series and the 13 financial features are outvoted.

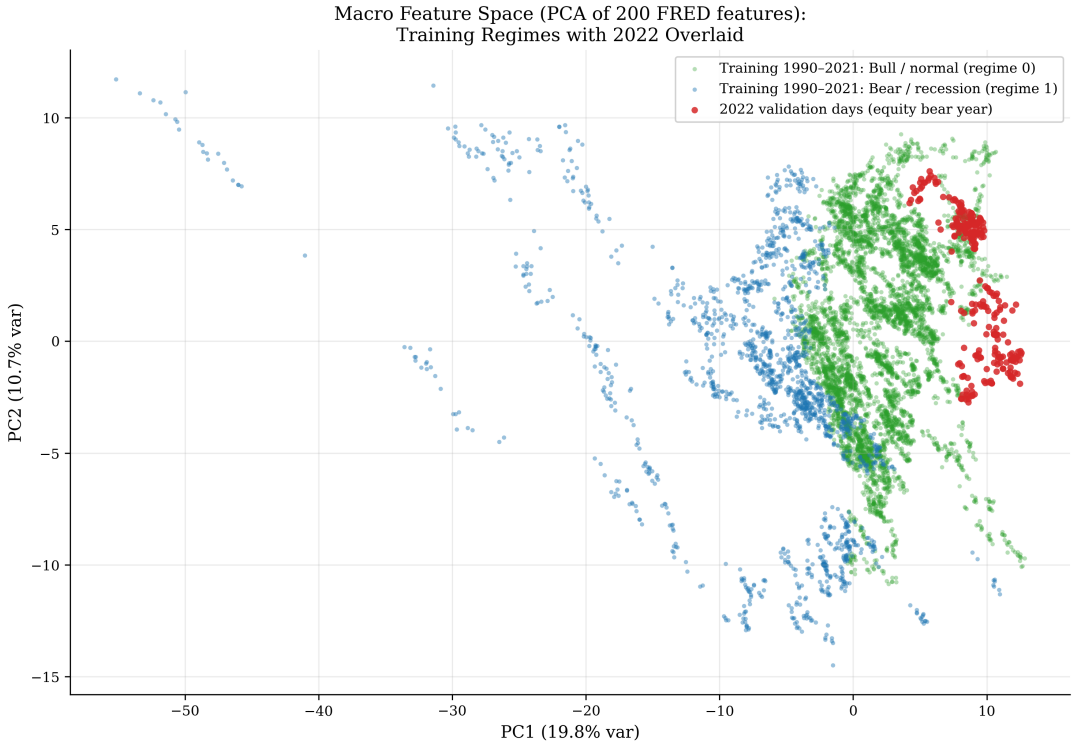



**Figure 1:** Why macroeconomic features miss the 2022 bear market. The macro feature space projected onto its first two principal components (fitted on the 1990 to 2021 training data, for visualisation only). Green marks training days decoded as Bull/normal. Blue marks training days decoded as Bear/recession (the 1998, 2001 to 2003, 2008 to 2009, and 2020 episodes). Red marks the 2022 validation days. The 2022 equity bear year sits inside the normal region, at its far edge, so the macro model reports no stress and assigns every validation day to Bull.

## 4.5 Candlestick Patterns and the Strategy Library

We detect eight canonical Japanese candlestick patterns on daily OHLC bars, four bullish and four bearish. A reversal pattern only means something after a move the other way, so every pattern has two preconditions. The first is a trend check. A bullish pattern counts only after a downtrend (the previous close below its 20 day simple moving average) and a bearish pattern only after an uptrend (the previous close above it). The second is a Bollinger Band filter (20 days, 2 standard deviations) carried over from the original implementation, which requires the close to be at or above 90% of the upper band or at or below the lower band.

**Bullish patterns** open a long and close a short.

1. **Hammer**. A small real body at the top of the trading range with a long lower shadow ($\geq 2\times$ the body) and almost no upper shadow.
2. **Dragonfly Doji**. Open $\approx$ close $\approx$ high (body $\leq 10\%$ of the range) with a lower shadow of at least 60% of the range, showing that buyers pushed price back up from the low.
3. **Bullish Engulfing**. A bearish candle followed by a larger bullish candle whose body fully engulfs the prior body.
4. **Inverted Hammer**. A small body at the bottom of the range with a long upper shadow ($\geq 2\times$ the body). Because the shape is a weaker reversal signal, it counts only when the close is at or below the lower Bollinger Band.

**Bearish patterns** close a long and open a short.

5. **Shooting Star**. The Inverted Hammer's shape after an advance, with a small body at the bottom and a long upper shadow.
6. **Gravestone Doji**. Open $\approx$ close $\approx$ low with an upper shadow of at least 60% of the range.
7. **Bearish Engulfing**. A bullish candle followed by a larger bearish candle whose body fully engulfs the prior body.
8. **Hanging Man**. The Hammer's shape after an advance, with a small body at the top and a long lower shadow.

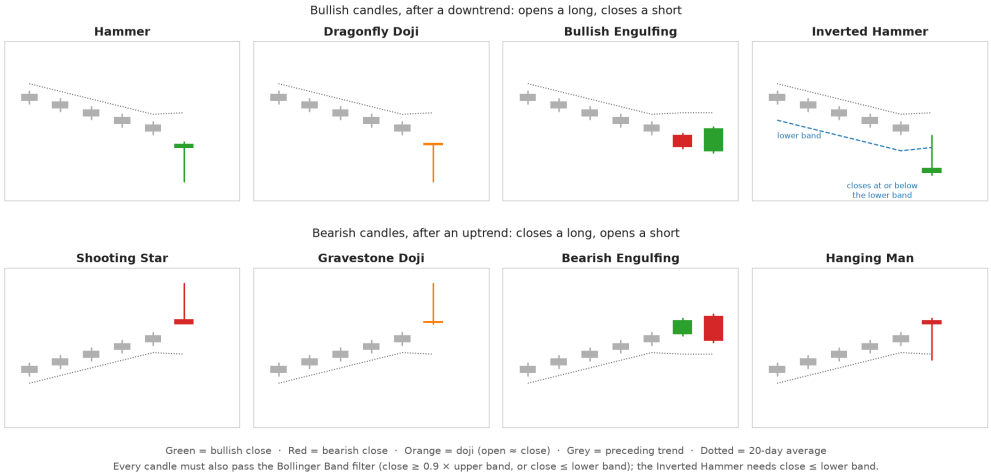



**Figure 2:** The eight candlestick patterns and their roles. The top row shows the bullish patterns, which are valid after a downtrend and open a long and close a short. The bottom row shows the bearish patterns, which are valid after an uptrend and close a long and open a short. Grey bars show the required preceding trend and the dotted line the 20 day average. Every pattern must also pass the Bollinger Band filter (close $\geq 0.9\times$ the upper band, or close $\leq$ the lower band), and the Inverted Hammer must close at or below the lower band. Bars are illustrative, not market data.

The detector uses standard body to shadow ratio thresholds and doji tolerance parameters following Nison [9] and Bulkowski [10], with no custom threshold tuning to avoid overfitting. At each daily bar, the detector is updated with the bar's OHLC values. If `detect()` returns at least one signal matching the current strategy's directional preference, an order is placed at the next open. When multiple patterns fire simultaneously, the highest strength signal is used as the entry label.

## 4.6 Strategy Design

### 4.6.1 Strategy Library

The library contains seven strategies (Table 6). Three are regime agnostic. They trade every candlestick signal and never consult the HMM. The other four apply the same signals but open positions only in a given regime, and close any open position when the regime changes. Each regime conditioned strategy is paired with the agnostic strategy that has the same long/short posture, which serves as its unconditional baseline. The regime for each day is the filtered (forward only) HMM label, so a decision on day $t$ uses only information available at the close of day $t$, and every order fills at the next day's open.

**Table 6:** Strategy Library

| Strategy | Trades in | Rule | Baseline |
|---|---|---|---|
| *Regime agnostic* | | | |
| `Agnostic_LongOnly` | Any regime | Every bullish candle opens a long, and it never shorts. | n/a |
| `Agnostic_ShortOnly` | Any regime | Every bearish candle opens a short, and it never buys. | n/a |
| `Agnostic_LongShort` | Any regime | Bullish candles open longs and bearish candles open shorts. | n/a |
| *Regime conditioned* | | | |
| `LongOnly_Bull` | Bull | Bullish candles open longs on Bull days, and longs close when the regime turns Bear. | `Agnostic_LongOnly` |
| `LongOnly_Bear` | Bear | A contrarian rule. Bullish candles open longs on Bear days (buy the dip), and longs close when the regime turns Bull. | `Agnostic_LongOnly` |
| `ShortOnly_Bear` | Bear | Bearish candles open shorts on Bear days, and shorts close when the regime turns Bull. | `Agnostic_ShortOnly` |
| `LongBull_ShortBear` | Bull / Bear | Longs on Bull days and shorts on Bear days. Each side closes when the regime flips. | `Agnostic_LongShort` |

All seven strategies are built from the same components, namely the candlestick detector, trend filter, Bollinger Band filter, position size, trailing stop, and profit take. They are nonetheless independent strategies, and each one is optimised independently (Section 4.6.2), so the tuned values of the Bollinger settings, trailing stop, and profit take differ from strategy to strategy (Table 7). A regime conditioned strategy therefore differs from its baseline in the regime gate and in the parameters that were tuned for it under that gate. Each strategy is run on all three universes, giving $7 \times 3 = 21$ (strategy, universe) combinations.

### 4.6.2 Position Sizing and Parameter Optimisation

**Fixed settings.** Position size is not optimised. Every new position is sized at 5% of current equity in all seven strategies, so that the strategies are compared under identical exposure and none of them is limited

by available cash. The Bollinger Band width is likewise held at 1.5 standard deviations.

**Optimisation window.** We first optimised over January 2022 to December 2023, chosen for symmetry with the two year evaluation period. Roughly half of that window, however, is the 2022 bear market. Optimising on it would tune the strategies to bear conditions far more heavily than those conditions are represented in practice, which is neither optimal nor realistic. We therefore extended the optimisation window to the full period from January 2021 to December 2023, so that Optuna tunes on a more representative mix of market conditions (the 2021 advance, the 2022 decline, and the 2023 recovery). Because the HMM is trained through December 2021, regime labels for 2021 are in sample for the HMM. They are used only for strategy tuning, never for evaluation.

**Rolling walk forward.** Parameters are tuned with a 28 fold rolling walk forward analysis over January 2021 to December 2023 (Figure 3). Each fold consists of a six month training window, on which parameters are chosen, followed immediately by a three month test window, on which those parameters are evaluated out of sample. Successive folds slide both windows forward by one month while keeping their lengths fixed, so every fold trains on the same amount of history and the oldest month drops out as a new one enters. The first fold trains on January to June 2021 and tests on July to September 2021. The last fold trains on April to September 2023 and tests on October to December 2023, the final quarter of the optimisation window, which gives 28 folds in total. Because the step (one month) is shorter than the test window (three months), consecutive test windows overlap, so the 28 out of sample results are averaged per strategy but are not independent samples.

**Objective and trials.** Optimisation uses a single objective. Within each training window an Optuna [27] study with its default Tree structured Parzen Estimator (TPE) sampler maximises the Sharpe ratio of the training backtest, computed from daily returns and annualised by $\sqrt{252}$. Each study runs 50 trials, and every trial is one full backtest of the strategy over the six month training window with the trial's candidate parameters. A trial must close at least two trades per month of training data to be scored. The best scoring trial's parameters are then run once on the fold's three month test window. If no trial in a fold meets the trade minimum, the strategy sits out that fold's test window (flat, zero return) rather than trade untrusted settings.

**Number of backtests.** For one strategy in one universe, each search stage runs $28$ folds $\times$ $50$ trials $=$ $1{,}400$ training backtests plus 28 out of sample test backtests. Across the seven strategies this is 9,800 training backtests per universe, and across the three universes 29,400 training backtests and 588 out of sample tests per stage. The two search stages described below together run 58,800 training backtests.

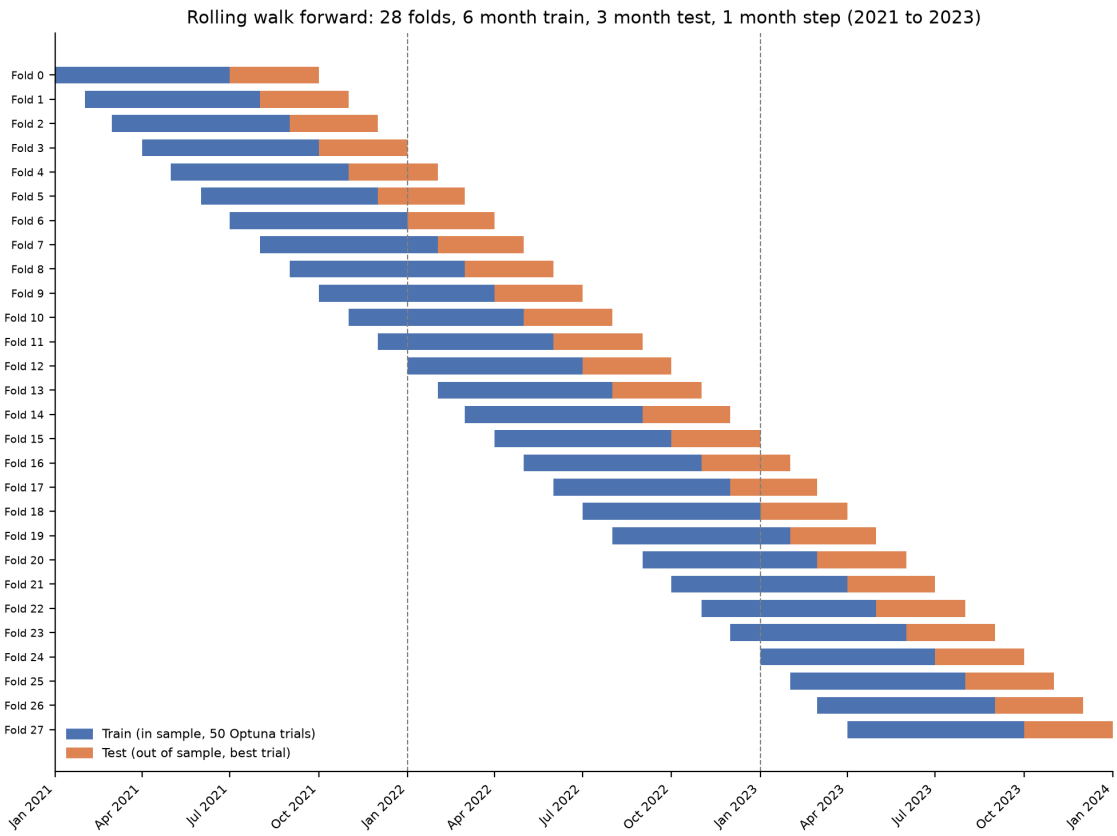



**Figure 3:** The 28 fold rolling walk forward used for parameter tuning, January 2021 to December 2023. Each row is one fold. Optuna runs 50 trials on the six month training window (blue), and the best trial's parameters are tested once on the following three month window (orange). Each fold slides one month later than the one above it. Dashed lines mark calendar year boundaries.

**Parameters.** Table 7 lists the parameters and their search ranges. The trailing stop closes a position once price moves the given fraction against its best close since entry, and the profit take closes it once it is up the given fraction from entry. The two band multipliers set how close to a Bollinger Band a candle must close to count. A long entry (or short exit) requires close $\leq f_L \times$ lower band, and a long exit (or short entry) requires close $\geq f_U \times$ upper band.

**Table 7:** Optimised and Fixed Parameters

**(a) Search setup**

| Parameter | Search range (step) | Stage 1 | Stage 2 |
|---|---|---|---|
| Trailing stop | 2% to 10% (0.5%) | Searched | Searched |
| Profit take | 3% to 20% (1%) | Searched | Searched |
| Bollinger period (days) | 60 to 100 (5) | Searched | Fixed from Stage 1 |
| Lower band multiplier $f_L$ | 1.00 to 1.10 (0.01) | Searched | Fixed from Stage 1 |
| Upper band multiplier $f_U$ | 0.80 to 1.00 (0.02) | Searched | Fixed from Stage 1 |
| Position size | Fixed at 5% of equity | Fixed | Fixed |
| Bollinger width | Fixed at 1.5 SD | Fixed | Fixed |
| Trend period (days) | Fixed at 60 | Fixed | Fixed |

**(b) Final values per strategy**

| | Stage 1 (held fixed in Stage 2) | | | Stage 2 | |
|---|---|---|---|---|---|
| **Strategy** | **BB period** | $f_L$ | $f_U$ | **Trailing stop** | **Profit take** |
| Agnostic_LongOnly | 80 | 1.02 | 0.90 | 3.5% | 6.0% |
| Agnostic_LongShort | 80 | 1.06 | 0.94 | 5.5% | 7.0% |
| Agnostic_ShortOnly | 85 | 1.05 | 0.94 | 6.0% | 7.0% |
| LongOnly_Bull | 75 | 1.07 | 0.88 | 4.0% | 3.0% |
| LongOnly_Bear | 80 | 1.02 | 0.92 | 3.5% | 7.0% |
| LongBull_ShortBear | 75 | 1.07 | 0.94 | 4.0% | 8.0% |
| ShortOnly_Bear | 80 | 1.04 | 0.92 | 5.0% | 13.5% |

Final values are the medians of each fold's best trial across the 28 folds and the three universes (LMC, SMC, COMBINED), rounded to the search grid. Stage 1 medians set the Bollinger settings held fixed in Stage 2, and Stage 2 medians set the trailing stop and profit take.

**Two stage search.** Stage 1 searches all five parameters together. The Bollinger period and the two band multipliers are related to each other, since together they decide how close to a band a candle must close before it counts as a signal. A longer period gives a smoother band, and the multipliers then set the trigger distance from that band, so these three are best chosen jointly. The trailing stop and profit take, by contrast, varied too much from fold to fold in Stage 1 to be pinned down while the band settings were also moving. We therefore used Stage 1 to optimise the three related Bollinger settings and fixed them at their medians, and in Stage 2 reoptimised only the trailing stop and profit take.

**Interactive example.** The full Stage 1 walk forward on the COMBINED universe (all five parameters searched, all seven strategies, every fold's training and test results) can be explored in the browser: COMBINED universe, five parameter walk forward (interactive).

Figure 4 shows why five free parameters is too many for a six month training window. It plots, for each search stage, the gap between the best trial's in sample Sharpe ratio and the Sharpe ratio the same parameters achieve out of sample. In Stage 1 the best trial reaches a mean in sample Sharpe ratio of 1.71 but only 0.28 out of sample, a gap of 1.43. With two free parameters in Stage 2 the in sample Sharpe falls to 1.19 while the out of sample Sharpe rises to 0.34, and the gap shrinks to 0.85. The gap narrows in every universe, from 1.25 to 0.81 on COMBINED, from 1.34 to 0.71 on LMC, and from 1.73 to 1.06 on SMC. With five parameters the optimiser has enough freedom to fit the noise of each training window, and fixing the three related band settings removes most of that freedom.

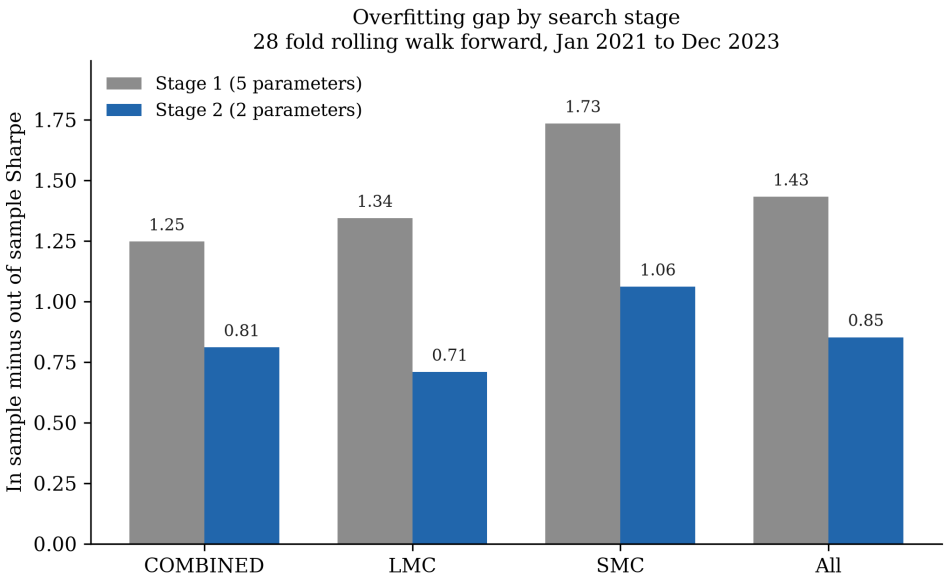



**Figure 4:** Gap between in sample and out of sample Sharpe ratio in the rolling walk forward, Stage 1 (all five parameters searched) versus Stage 2 (Bollinger settings fixed, two parameters searched). Each bar is the mean over the folds of all seven strategies of the best trial's training window Sharpe minus the same parameters' test window Sharpe. All pools the three universes. Folds in which the strategy sat out or made no test trades are excluded (527 of 588 in Stage 1, 506 in Stage 2).

## 4.7 Phase Lifecycle

The end to end system runs in three sequential stages, illustrated in Figure 5. Phase 1 trains and freezes the regime model on 2016 to 2021 data and turns it into a daily filtered regime label. Phase 2 tunes every (strategy, universe) combination on 2021 to 2023 with the two stage rolling walk forward. The final parameters are then run once on the locked 2024 to 2025 evaluation period, which produces every result reported in Section 6.

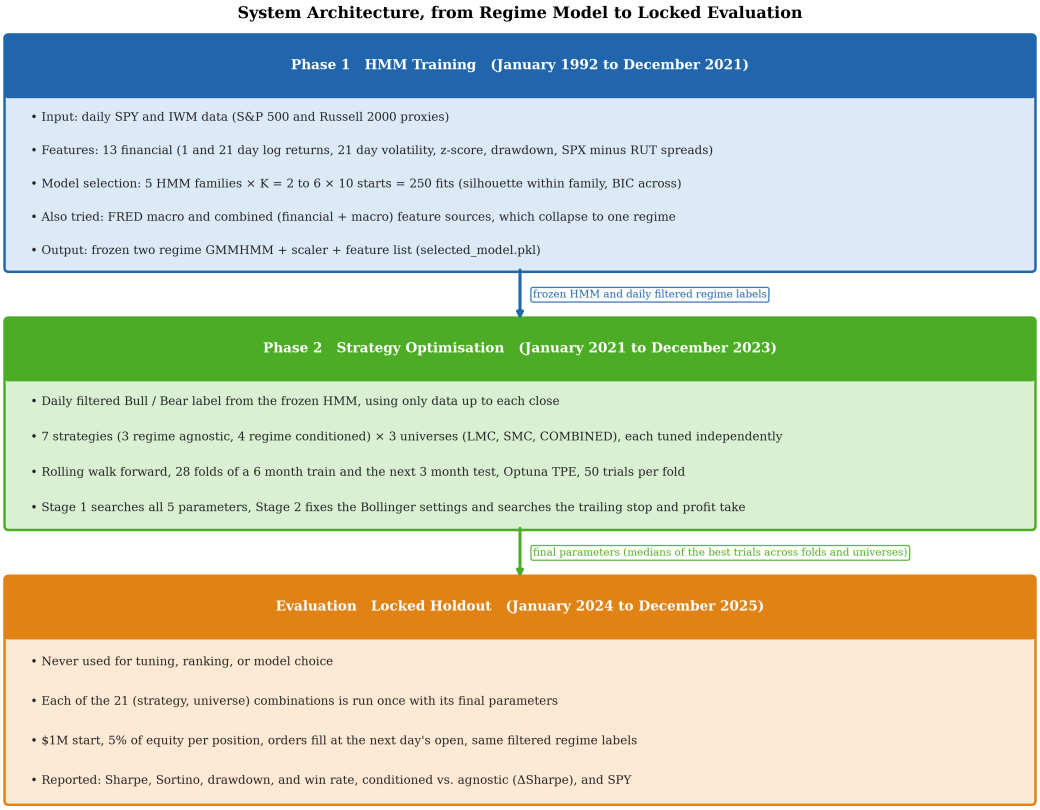



**Figure 5:** System architecture. Phase 1 trains the HMM on historical data, freezes it, and produces the daily filtered regime labels. Phase 2 tunes each strategy configuration independently with the two stage rolling walk forward over the optimisation period. The final parameters are then run once on the locked evaluation period, which is never used for tuning or model choice.

### 4.7.1 Phase 1 (HMM Training)

Phase 1 ingests six years of financial market data (2016 to 2021). After feature engineering, the GMMHMM is trained via Baum-Welch. From the candidates in Sections 4.2.4 and 4.4 we carry forward the two regime `GMMHMM_diag_mix3` trained on the 13 financial features. Every model family chose two states, and the financial model is the only feature source that produces a realistic split between Bull and Bear days, whereas the macro and combined models place almost every day in a single regime. The trained model, `StandardScaler`, and feature column list are serialised to a pickle file (`selected_model.pkl`). This artifact is frozen and never modified after Phase 1 completes.

The frozen model is then run once over every trading day, and the labels are saved to a daily regime file that all backtests read. For each day $D$ the model is applied to the features up to and including $D$, and the regime with the highest probability on that last day is kept. This filtered label uses only information available at the close of $D$, which is what a live system would know that evening. It differs from the Viterbi path of Section 4.2.3, which decodes a whole period at once, so that the label of day $D$ also depends on later days. A strategy reads the filtered regime of day $D$ after the close, and any order it places fills at the next day's open. The regime that governs each day's trading is therefore always computed from data up to the previous close.

### 4.7.2 Phase 2 (Strategy Optimisation)

Phase 2 is split into two nonoverlapping subperiods.

- **Optimisation (OPT).** January 2021 to December 2023. Used for the two stage rolling walk forward Optuna search (Section 4.6.2), which first sets the Bollinger period and the two band multipliers

and then the trailing stop and profit take of each strategy.

- **Evaluation (EVAL).** January 2024 to December 2025. A locked holdout period used exclusively for reporting final performance metrics. No parameter selection, strategy ranking, or fitting decisions use data from this period.

For each of the 7 strategies $\times$ 3 universes = 21 (strategy, universe) combinations, the backtester simulates daily decisions with the final parameters of Table 7. After each close the strategy works through every stock in its universe in the following order.

1. **Regime.** It reads the day's filtered regime from the saved Phase 1 file and checks whether the strategy may hold a long or a short in that regime. The regime agnostic strategies skip this check.
2. **Open positions.** If the stock is already held, the position is closed by the first exit rule that applies. These are an opposite candlestick pattern with the close near the opposite Bollinger Band, the trailing stop, the profit take, and finally a regime in which the strategy may no longer hold that side. A long closes on a bearish pattern when the close is at or above $f_U\times$ the upper band, and a short closes on a bullish pattern when the close is at or below $f_L\times$ the lower band.
3. **New positions.** If the stock is not held, a position is opened only when three conditions hold together. A candlestick pattern must appear after the required trend, the close must be near the matching Bollinger Band, and the day's regime must allow that side. A long needs a bullish pattern with the close at or below $f_L\times$ the lower band, and a short needs a bearish pattern with the close at or above $f_U\times$ the upper band. Each new position is sized at 5% of current equity.

Orders placed after the close fill at the next day's open, so no decision relies on a price that was not yet known.

### 4.8 Universe Selection

The universes are carried over from our earlier QuantConnect algorithm, which placed first and third in two separate quarterly competitions. That algorithm traded a large market cap (LMC) universe and a small market cap (SMC) universe. We keep both unchanged and add their union (COMBINED), so the regime conditioned strategies are tested on the same stocks as the system they build on. Membership of each universe is fixed at a January 2022 snapshot and is never revised with later information, so no stock enters a universe because of how it performed afterwards. A few small cap names from the original list were later acquired or no longer have a usable price history, and these are left out of the backtests. The SMC universe therefore carries a mild survivorship bias.

## 5 Experimental Setup

### 5.1 Training, Validation, and Test Splits

The pipeline uses the three data periods listed below. HMM model selection (silhouette and BIC) is computed on the training period itself, so it is in sample, and the optimisation window shares 2021 with the HMM training window. The evaluation period overlaps neither.

**Table 8:** Dataset Splits

| Split | Period | Purpose |
|---|---|---|
| Training | Jan 2016 to Dec 2021 | HMM parameter estimation (Baum-Welch) and model selection (silhouette, BIC) |
| Strategy Optimisation | Jan 2021 to Dec 2023 | Rolling walk forward Optuna tuning |
| Evaluation | Jan 2024 to Dec 2025 | Locked holdout for final performance reporting |

The 2024 to 2025 evaluation period is never used for strategy ranking, parameter selection, or any fitting decision. All reported evaluation period metrics are therefore free of in sample bias.

### 5.2 Evaluation Metrics

We evaluate strategies using the following metrics.

$$\text{Sharpe} = \frac{\bar{r}_d}{\sigma_d} \cdot \sqrt{252} \tag{11}$$

$$\text{Sortino} = \frac{\bar{r}_d}{\sigma_d^-} \cdot \sqrt{252}, \quad \sigma_d^- = \text{std of negative daily returns} \tag{12}$$

$$\text{Max Drawdown} = \min_t \frac{E_t - \max_{s \le t} E_s}{\max_{s \le t} E_s} \tag{13}$$

$$\text{Calmar} = \frac{\text{Annualised Return}}{|\text{Max Drawdown}|} \tag{14}$$

$$\text{Profit Factor} = \frac{\sum_{i: r_i > 0} r_i}{\left| \sum_{i: r_i \le 0} r_i \right|} \tag{15}$$

where $E_t$ is portfolio equity at time $t$, $r_i$ is trade $i$ PnL, and $\bar{r}_d$, $\sigma_d$ are the mean and standard deviation of daily returns. All annualisation uses $\sqrt{252}$.

## 6 Backtest Results

### 6.1 Regime Classification

The two regime HMM trained on the financial feature set produces geometrically well separated clusters (silhouette $= 0.317$) in the 13 dimensional feature space. The regime timeline (Figure 7) shows persistent regime assignments over the validation and evaluation periods, with the 2022 bear market (high VIX, negative SPY returns) consistently assigned to Regime 1 (Bear), and the 2023 to 2024 recovery and subsequent bull run assigned to Regime 0 (Bull).

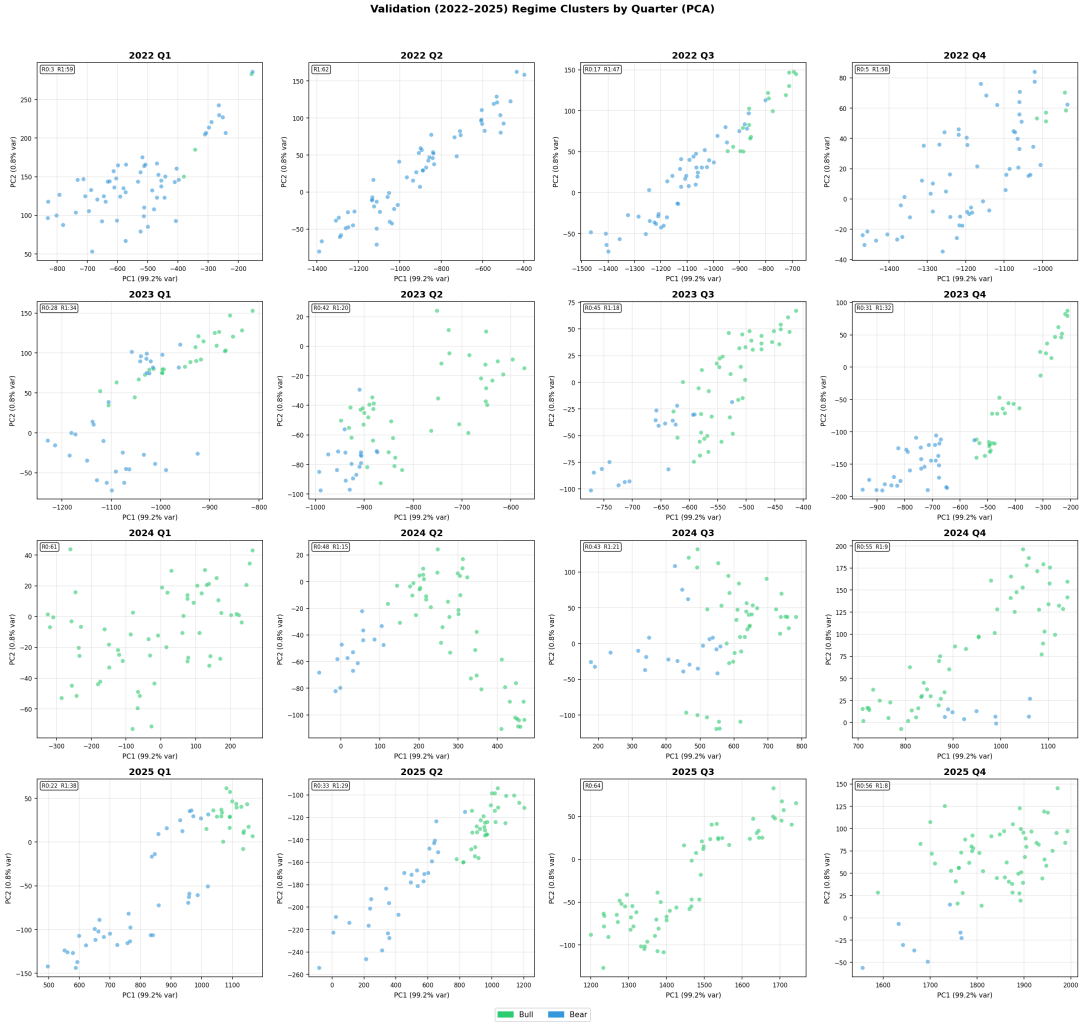



**Figure 6:** PCA projection of the 13 dimensional financial feature space, colour coded by HMM assigned regime. The two regimes are geometrically well separated (silhouette $= 0.317$).

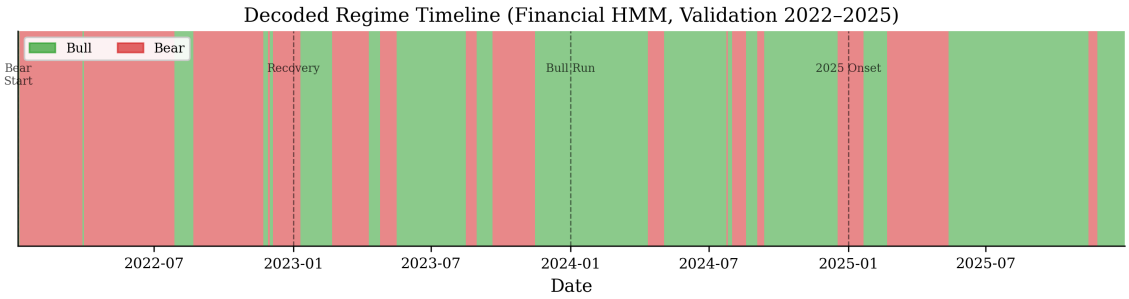



**Figure 7:** Market regime timeline from the two state financial HMM (January 2022 to December 2025). Colour bands correspond to Bull (Regime 0) and Bear (Regime 1). The 2022 bear market, 2023 recovery, and 2024 to 2025 bull run are clearly delineated.

## 6.2 Strategy Performance

Table 9 reports all 21 (strategy, universe) combinations, the seven strategies of Table 6 on each of the three universes, run with the final parameters of Table 7 over the evaluation period (January 2024 to December 2025). Appendix A gives the full set of metrics.

**Table 9:** Strategy Performance, Evaluation Period (January 2024 to December 2025)

| Strategy | Ret. | Sortino | Sharpe | MDD | Win% |
|---|---|---|---|---|---|
| **COMBINED (LMC + SMC)** | | | | | |
| *Regime agnostic* | | | | | |
| Agnostic_LongOnly | +5.4% | 0.97 | 0.60 | −3.8% | 46.7% |
| Agnostic_LongShort | −6.1% | −0.47 | −0.33 | −12.1% | 42.4% |
| Agnostic_ShortOnly | −18.3% | −1.57 | −1.09 | −21.0% | 36.2% |
| *Regime conditioned* | | | | | |
| LongOnly_Bull | **+5.8%** | **1.52** | **1.01** | **−2.5%** | **54.8%** |
| LongOnly_Bear | +4.7% | 1.01 | 0.60 | −3.9% | 53.3% |
| LongBull_ShortBear | +3.7% | 0.60 | 0.39 | −5.9% | 45.1% |
| ShortOnly_Bear | −2.4% | −0.43 | −0.28 | −10.0% | 38.8% |
| **LMC (large cap)** | | | | | |
| *Regime agnostic* | | | | | |
| Agnostic_LongOnly | **+6.1%** | 1.80 | 1.11 | −1.9% | 52.2% |
| Agnostic_LongShort | −9.9% | −1.17 | −0.85 | −12.4% | 39.9% |
| Agnostic_ShortOnly | −18.5% | −2.26 | −1.62 | −19.4% | 33.1% |
| *Regime conditioned* | | | | | |
| LongOnly_Bull | +4.9% | **2.10** | **1.32** | −2.2% | 58.3% |
| LongOnly_Bear | +3.9% | 1.39 | 0.85 | **−1.9%** | **61.9%** |
| LongBull_ShortBear | +3.4% | 0.90 | 0.52 | −4.3% | 45.1% |
| ShortOnly_Bear | −1.7% | −0.49 | −0.29 | −6.0% | 34.0% |
| **SMC (small cap)** | | | | | |
| *Regime agnostic* | | | | | |
| Agnostic_LongOnly | −0.7% | −0.20 | −0.13 | −3.4% | 40.0% |
| Agnostic_LongShort | −2.4% | −0.27 | −0.18 | −10.6% | 44.5% |
| Agnostic_ShortOnly | −1.5% | −0.19 | −0.13 | −6.0% | 41.2% |
| *Regime conditioned* | | | | | |
| LongOnly_Bull | **+0.9%** | **0.35** | **0.24** | **−2.2%** | **50.5%** |
| LongOnly_Bear | +0.8% | 0.32 | 0.19 | −2.7% | 42.4% |
| LongBull_ShortBear | +0.3% | 0.09 | 0.07 | −6.1% | 45.0% |
| ShortOnly_Bear | −0.6% | −0.18 | −0.13 | −5.4% | 49.0% |

Ret. is the total return over the two years. Sortino and Sharpe use daily equity, a zero risk free rate, and $\sqrt{252}$ annualisation. MDD is the maximum drawdown. Win% is the share of closed round trips with a positive net P&L. Starting capital \$1,000,000, 5% of equity per position. Regime conditioned strategies are shaded, and the best value of each column in each universe is in bold.

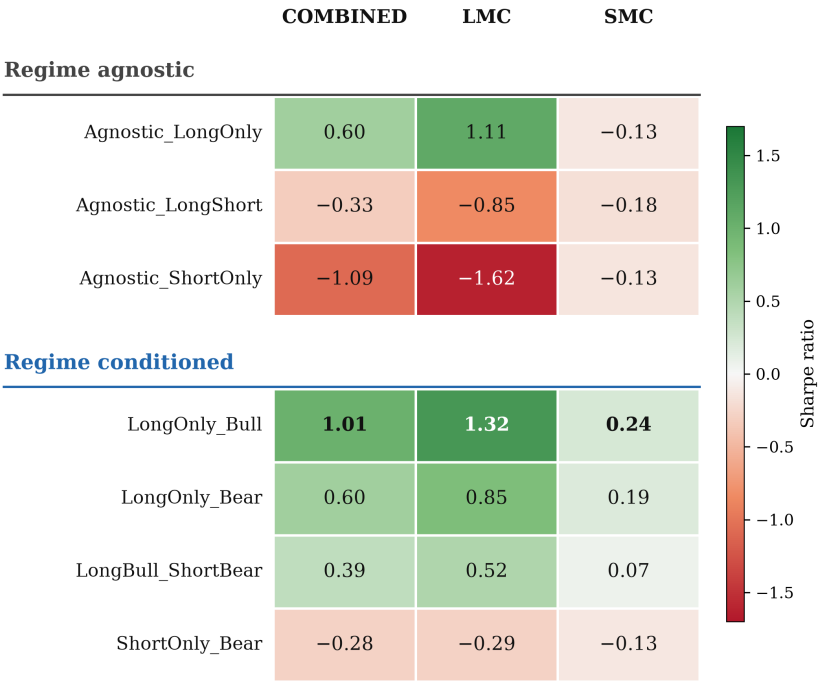



**Figure 8:** Sharpe ratio heatmap across the 21 (strategy, universe) combinations over the evaluation period (January 2024 to December 2025), using the final parameters of Table 7 (values from Table 9). The top block holds the regime agnostic strategies and the bottom block the regime conditioned strategies. Green is a positive Sharpe ratio and red a negative one. Bold marks the best strategy in each universe.

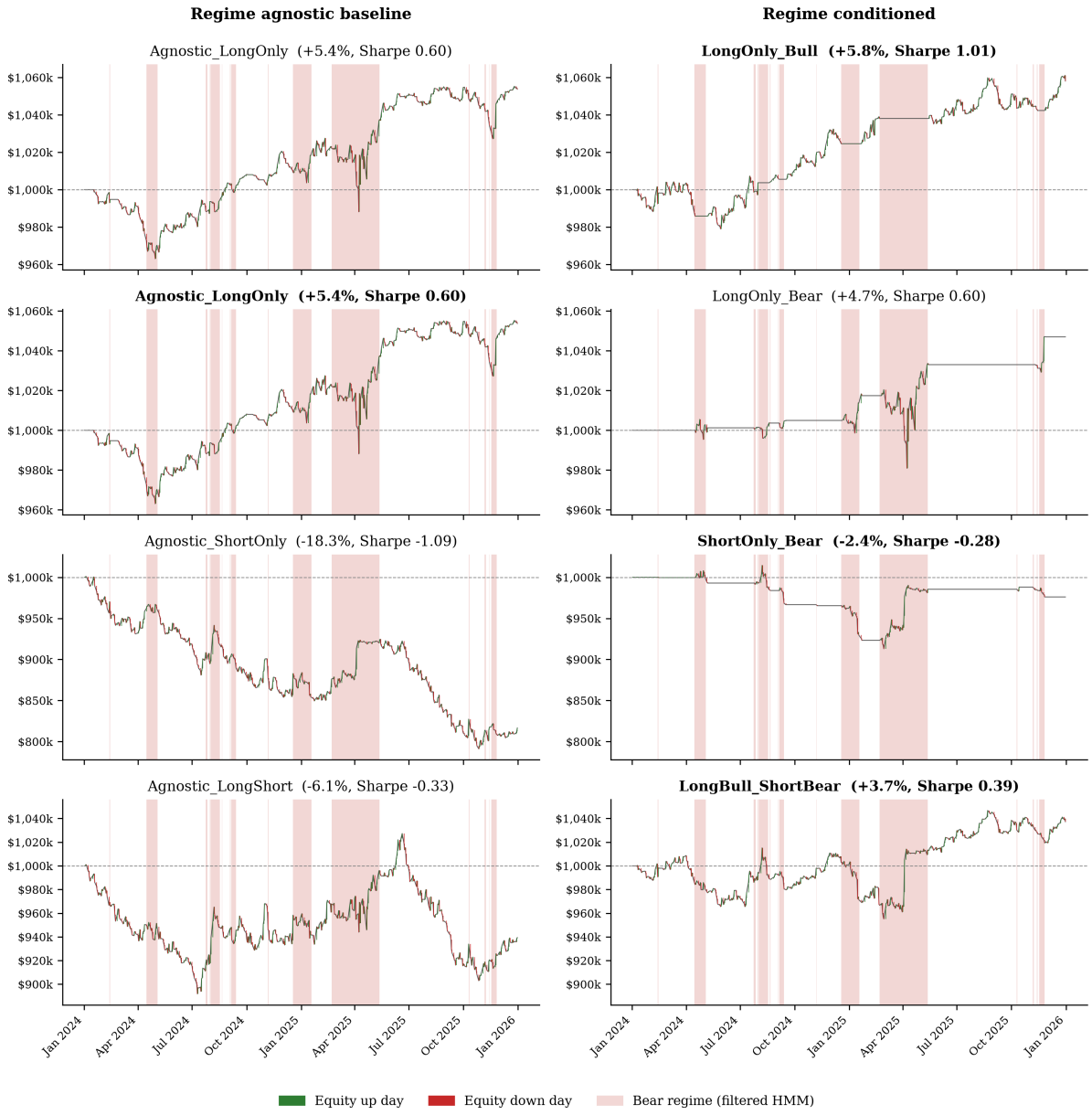



**Figure 9:** Daily equity candlesticks of each regime conditioned strategy (right) next to its regime agnostic baseline (left), COMBINED universe, evaluation period (January 2024 to December 2025). Each row is one pair from Table 6, and both panels in a row share the same vertical scale. Shading marks Bear days from the filtered HMM label, and the dashed line is the $1,000,000 starting capital. Flat stretches are days with no open positions. Titles give the total return and Sharpe ratio over the period, and the higher Sharpe in each pair is in bold. In the LongOnly_Bear row the Sharpe ratios tie at two decimals, so the higher total return (Agnostic_LongOnly) is in bold.

## 6.3 Regime Conditioned Strategies vs. Agnostic Baselines

Each regime conditioned strategy is compared with the regime agnostic strategy of the same long/short posture (Table 6). The two are similar strategies built from the same mechanics, but they are not identical. Each is optimised independently, so besides the regime gate they also differ in their tuned Bollinger settings, trailing stop, and profit take (Table 7). The difference in Sharpe ratio, $\Delta$Sharpe, therefore measures the effect of conditioning on the HMM regime together with the parameters tuned under that gate, rather than the gate alone. Table 10 reports $\Delta$Sharpe for each pair on each universe over the evaluation period, using the Sharpe ratios of Table 9. A positive $\Delta$Sharpe means the regime conditioned

strategy improved on its baseline.

**Table 10:** ΔSharpe of Regime Conditioned Strategies vs. Their Agnostic Baselines, Evaluation Period (January 2024 to December 2025)

| **Strategy** (regime conditioned) | **Baseline** | **Strategy Sharpe** | **Baseline Sharpe** | **ΔSharpe** |
|---|---|---|---|---|
| **COMBINED (LMC + SMC)** | | | | |
| LongOnly_Bull | Agnostic_LongOnly | **1.01** | 0.60 | +0.41 |
| LongOnly_Bear | Agnostic_LongOnly | *0.60* | *0.60* | +0.00 |
| ShortOnly_Bear | Agnostic_ShortOnly | **−0.28** | −1.09 | +0.81 |
| LongBull_ShortBear | Agnostic_LongShort | **0.39** | −0.33 | +0.72 |
| **LMC (large cap)** | | | | |
| LongOnly_Bull | Agnostic_LongOnly | **1.32** | 1.11 | +0.21 |
| LongOnly_Bear | Agnostic_LongOnly | 0.85 | **1.11** | −0.26 |
| ShortOnly_Bear | Agnostic_ShortOnly | **−0.29** | −1.62 | +1.33 |
| LongBull_ShortBear | Agnostic_LongShort | **0.52** | −0.85 | +1.37 |
| **SMC (small cap)** | | | | |
| LongOnly_Bull | Agnostic_LongOnly | **0.24** | −0.13 | +0.37 |
| LongOnly_Bear | Agnostic_LongOnly | **0.19** | −0.13 | +0.32 |
| ShortOnly_Bear | Agnostic_ShortOnly | *−0.13* | *−0.13* | +0.00 |
| LongBull_ShortBear | Agnostic_LongShort | **0.07** | −0.18 | +0.25 |

ΔSharpe is the regime conditioned Sharpe minus its baseline's. In each row the higher of the two Sharpe ratios is in bold, and ties at two decimals are in script.

## 6.4 Candlestick Pattern Effectiveness by Regime

Figure 10 shows the win rate of each candlestick pattern in the Bull and Bear regimes of the two state HMM over the evaluation period. Trades from all seven strategies are used (COMBINED universe, 1,952 round trips). Each strategy runs its own portfolio with its own optimised parameters, so its trades are counted separately even when two strategies act on the same signal. Each trade is matched to the candle that opened it and assigned the filtered HMM regime of that signal day. A win is a round trip with a positive net P&L.

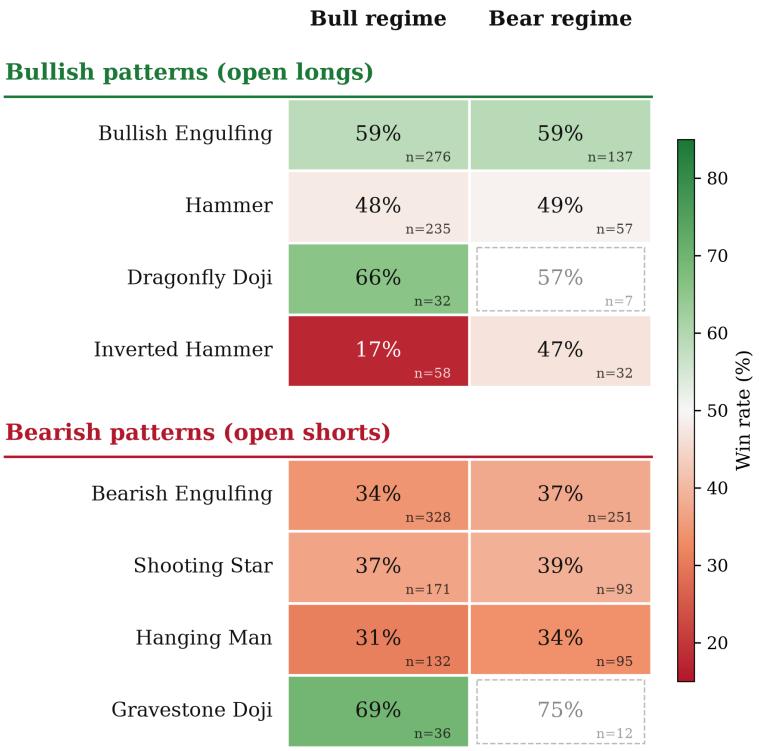



**Figure 10:** Candlestick pattern win rates by market regime (Bull and Bear), all seven strategies, COMBINED universe, January 2024 to December 2025. Each cell gives the win rate, with the number of trades ($n$) in the corner. Green is above 50% and red is below. Uncoloured cells with a dashed outline have fewer than 30 trades and should be read with caution.

The clearest result is the Bullish Engulfing pattern, which wins 59% of the time in both regimes and does so on a large sample (276 trades in Bull and 137 in Bear). It is the only pattern that is both reliably profitable and frequent, and its win rate does not depend on the regime. The Dragonfly Doji also does well in the Bull regime (66%, 32 trades), while its Bear cell has too few trades to judge. The Hammer is close to a coin flip in both regimes (48% and 49%), and the Inverted Hammer is the weakest long signal, winning only 17% of its 58 Bull regime trades.

On the short side the picture is consistently poor. Bearish Engulfing, Shooting Star, and Hanging Man win between 31% and 39% of the time in both regimes, and together they account for 1,070 trades with a pooled win rate of 35%. The one exception is the Gravestone Doji, which wins 69% of its 36 trades in the Bull regime. Overall, the 1,118 short trades opened on a bearish pattern win 37% of the time, against 52% for the 834 long trades opened on a bullish pattern. Shorting on a candlestick reversal does not work well in either regime over this period, which is consistent with the weak results of the short strategies in Table 9.

## 6.5 Small Cap vs. Large Cap Analysis

Figure 11 compares the average trade return of the large cap (LMC) and small cap (SMC) universes in the Bull and Bear regimes, testing whether the small cap effect depends on the market regime. It uses the round trips of all seven strategies over the evaluation period, as in the pattern analysis of Figure 10, with each trade assigned the filtered regime of the signal day that opened it. Trades are split by regime and by side, since longs and shorts behave very differently.

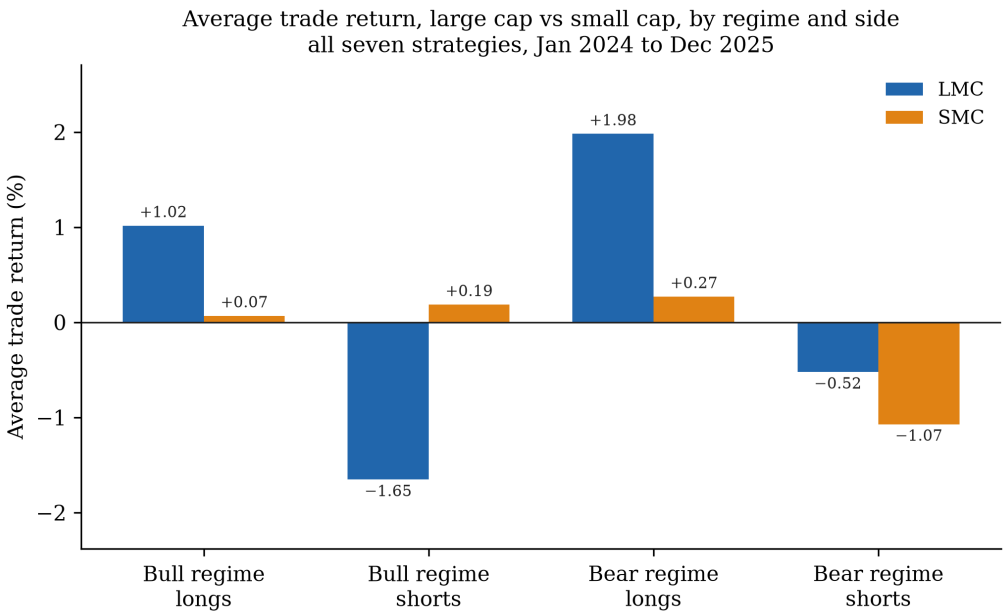



**Figure 11:** Average trade return (%) for LMC (large cap) vs. SMC (small cap) by regime and side, all seven strategies, January 2024 to December 2025. Labels give the mean return per trade. Trade counts (LMC, SMC) are 340 and 290 for Bull longs, 456 and 272 for Bull shorts, 132 and 107 for Bear longs, and 299 and 146 for Bear shorts.

Large caps respond much more strongly to the direction of the trade. LMC longs earn $+1.02\%$ per trade in the Bull regime and $+1.98\%$ in the Bear regime, against only $+0.07\%$ and $+0.27\%$ for SMC longs. The same sensitivity works against LMC shorts, which lose $-1.65\%$ per trade in the Bull regime, the worst result in the figure. Small caps are flatter in both directions, with small average gains and losses except for Bear regime shorts ($-1.07\%$). Shorts lose money in both universes and both regimes apart from a small gain on SMC in the Bull regime. The small cap universe therefore adds little on either side, and the edge of the long strategies comes mainly from large caps, especially when they buy Bear regime dips.

## 6.6 Benchmark Comparison

Table 11 compares the best regime conditioned strategy and its regime agnostic baseline against SPY buy and hold over the same period. The best regime conditioned strategy is `LongOnly_Bull` on the LMC universe, the highest Sharpe ratio in Table 9, and its baseline is `Agnostic_LongOnly` on the same universe. The period is the evaluation period, January 2024 to December 2025.

**Table 11:** Strategy vs. SPY Buy and Hold, Evaluation Period (January 2024 to December 2025)

| Metric | LongOnly_Bull (LMC) | Agnostic_LongOnly (LMC) | SPY |
|---|---|---|---|
| Total Return | +4.9% | +6.1% | +47.8% |
| Annualised Return | +2.4% | +3.0% | +21.6% |
| Sharpe Ratio | **1.32** | 1.11 | 1.28 |
| Sortino Ratio | **2.10** | 1.80 | 1.91 |
| Max Drawdown | -2.2% | -1.9% | -18.8% |
| Calmar Ratio | 1.08 | **1.62** | 1.15 |

All metrics from daily equity (strategies) or daily adjusted close (SPY, dividends reinvested), zero risk free rate, annualised with $\sqrt{252}$. Calmar is the annualised return divided by the absolute maximum drawdown. The best value of each ratio is in bold.

Because every position is sized at 5% of equity, the strategies carry far less market exposure than a fully invested SPY position, so raw returns are not directly comparable. Risk adjusted measures (Sharpe, Sortino, Calmar) are the fairer comparison.

# 7 Discussion

## 7.1 Key Findings

Our results support five main conclusions.

**(1) A two regime financial model gives a usable market regime signal.** All five HMM families select two states, and the model trained on 13 financial features produces well separated regimes (silhouette $= 0.317$) with a realistic split between Bull and Bear days. The macro and combined feature sources collapse to a single regime and miss the 2022 bear market entirely, so more data does not by itself give better regime detection.

**(2) Conditioning on the regime improves similar strategies on a risk adjusted basis.** Over the 2024 to 2025 evaluation period the regime conditioned strategy has the higher Sharpe ratio than its agnostic counterpart in 9 of 12 comparisons, ties in 2, and trails in 1. `LongOnly_Bull` is the best strategy in every universe and reaches a Sharpe ratio of 1.32 on large caps, against 1.11 for `Agnostic_LongOnly` and 1.28 for SPY, with a maximum drawdown of only 2.2% against 18.8% for SPY. The largest improvements are on the short side ($+1.37$ and $+1.33$ on LMC), where the regime gate turns large losses into much smaller ones rather than into profits. Because each strategy is optimised independently, these differences reflect the regime gate together with the parameters tuned under it.

**(3) Candlestick reversals work on the long side and fail on the short side.** Bullish Engulfing wins 59% of its trades in both regimes on a large sample, and the Dragonfly Doji does well in the Bull regime. Bearish patterns are consistently weak, and short trades opened on a bearish pattern win 37% of the time against 52% for longs. The Gravestone Doji in the Bull regime is the only short signal with a high win rate.

**(4) Large caps carry the edge.** Long trades on large caps earn $+1.02\%$ per trade in the Bull regime and $+1.98\%$ in the Bear regime, against $+0.07\%$ and $+0.27\%$ on small caps. Large caps respond more strongly to the direction of the trade in both directions, so they also produce the largest short losses, while small caps add little either way.

**(5) A staged search generalises better than optimising everything at once.** Fewer free parameters reduce overfitting, but the more useful lesson is how to reduce them. The Bollinger period and the two band multipliers are parts of the same indicator and together decide when a candle is close enough to a band to count, whereas the trailing stop and profit take control the exits and act independently of the entry. We therefore searched all five parameters in a first stage, fixed the three related Bollinger settings at their medians, and searched only the trailing stop and profit take in a second stage. This cut the mean gap between in sample and out of sample Sharpe in the walk forward from 1.43 to 0.85, and the gap narrowed in every universe. Grouping related parameters, fixing them, and then optimising the remaining ones in a later stage is a simple two stage (or, more generally, multiple stage) alternative to a single search over every parameter, and it carries over better to unseen data.

## 7.2 What We Are Not Claiming

We do not claim that these strategies beat the S&P 500. Each position is sized at 5% of equity, so the strategies are only partly invested and their total returns are far below those of a fully invested SPY position (4.9% for the best strategy against 47.8% for SPY over 2024 to 2025). Position sizing matters a great deal in any comparison with the index. A practical system would also need a rule for days with no signal, for example holding SPY passively while the candlestick strategies are flat, and we do not test

such a combination here. The evaluation period also spans only two years and does not include a severe bear market. Bailey and López de Prado's [18] Deflated Sharpe Ratio framework suggests caution when interpreting Sharpe ratios from a 21 combination search without explicit multiple testing correction.

What we do claim is narrower. When a candlestick strategy is developed, the market regime has a clear effect on how it performs. Across the 12 comparisons between a regime conditioned strategy and its regime agnostic counterpart, the conditioned version has the higher Sharpe ratio in 9, ties in 2, and trails in only 1. Using an HMM, or another form of regime identification, to decide when a trading strategy should take part in the market can therefore be a useful addition to strategy design. Running the same candlestick patterns through many strategies also proved a useful way to see the strengths and weaknesses of the patterns themselves, both overall and by regime (Section 6.4). Bullish Engulfing, for example, holds up in both regimes, while the Inverted Hammer and most bearish patterns are weak whatever the regime.

### 7.3 Limitations

- **Short evaluation period.** The 2024 to 2025 evaluation spans only one market regime cycle. Results may not generalise across longer or different market cycles.
- **Transaction cost model.** A flat $1.00 commission is a simplification. Bid and ask spreads, market impact, and slippage (especially in SMC tickers) would reduce returns in production and were not stress tested beyond the flat commission.
- **Small cap liquidity.** Several SMC tickers have thin order books. The position size may not be achievable without significant price impact at production scale.
- **Multiple testing.** We evaluate 21 (strategy, universe) combinations. While the OPT/EVAL split mitigates in sample overfitting, the Deflated Sharpe Ratio [18] would further reduce the statistical significance of our best result.

### 7.4 Future Work

1. **Continuous regime inference.** Replacing hard filtered assignments with regime probabilities (soft assignments) as position sizing weights may smooth transitions and improve returns.
2. **Intraday candlestick signals.** The regimes would stay on a daily basis, while the candlestick strategies would trade on intraday bars (for example hourly or 5 minute candles). The daily regime would set the direction a strategy may take, and the intraday patterns would time the entries and exits within it.
3. **Optimising position size.** Every position is currently sized at a fixed 5% of equity. We did not optimise this percentage because the strategies trade often and across many stocks at once, so the size of each new position interacts with the cash already committed to the open ones. Letting the size vary, for example by signal strength, regime, or the number of open positions, is fairly complicated to build into a backtest, and adding it to the search would make the already large walk forward optimisation far more expensive to run. Treating position size as a tuned parameter is a natural next step.

## 8 Conclusion

We have presented a systematic empirical study of regime conditional candlestick pattern trading. A two state Gaussian Mixture HMM trained on 13 financial features identifies persistent Bull and Bear market regimes, and its filtered label, which uses only data up to each day's close, gates a library of seven candlestick strategies. Each strategy is tuned independently with a two stage rolling walk forward search,

and all of them are tested on a locked 2024 to 2025 evaluation period across large cap, small cap, and combined universes.

Conditioning on the regime improves similar strategies on a risk adjusted basis. The regime conditioned strategy has the higher Sharpe ratio than its agnostic counterpart in 9 of 12 comparisons, ties in 2, and trails in 1. The best strategy, `LongOnly_Bull` on large caps, reaches a Sharpe ratio of 1.32 against 1.11 for its agnostic counterpart and 1.28 for SPY, with a maximum drawdown of 2.2% against 18.8% for SPY. The edge comes from buying bullish candlestick reversals in large caps, while shorting on bearish patterns loses money in almost every setting. On the optimisation side, fixing the related Bollinger settings and tuning the exits in a second stage cuts the gap between in sample and out of sample Sharpe from 1.43 to 0.85.

These results come with clear limits. The evaluation period covers only two years, the regime conditioned gains on the short side mostly reduce losses rather than create profits, and with 5% of equity per position the strategies carry far less exposure than a fully invested index, so their total returns are small (4.9% for `LongOnly_Bull` on large caps against 47.8% for SPY). Practitioners should therefore treat regime detection as a useful structural prior that improves risk adjusted returns, rather than as a guaranteed source of alpha.

# A Full Strategy Results

Table 12 reports all 21 strategy and universe combinations (7 strategies $\times$ 3 universes) over the evaluation period (January 2024 to December 2025), using the final parameters of Table 7. All metrics are computed from daily equity with a zero risk free rate and $\sqrt{252}$ annualisation. Calmar is the annualised return divided by the absolute maximum drawdown. Rows are grouped by universe, and within each universe the regime conditioned strategies (shaded) follow their regime agnostic baselines. The best value of each column in each universe is in bold (for MDD, the shallowest drawdown).

**Table 12:** All 21 Strategy and Universe Combinations, Evaluation Period (January 2024 to December 2025)

| Strategy | Ret. | Ann. | Sharpe | Sortino | MDD | Calmar |
|---|---|---|---|---|---|---|
| **COMBINED (LMC + SMC)** | | | | | | |
| *Regime agnostic* | | | | | | |
| Agnostic_LongOnly | +5.4% | +2.6% | 0.60 | 0.97 | −3.8% | 0.69 |
| Agnostic_LongShort | −6.1% | −3.1% | −0.33 | −0.47 | −12.1% | −0.25 |
| Agnostic_ShortOnly | −18.3% | −9.6% | −1.09 | −1.57 | −21.0% | −0.46 |
| *Regime conditioned* | | | | | | |
| LongOnly_Bull | **+5.8%** | **+2.9%** | **1.01** | **1.52** | **−2.5%** | **1.14** |
| LongOnly_Bear | +4.7% | +2.3% | 0.60 | 1.01 | −3.9% | 0.60 |
| LongBull_ShortBear | +3.7% | +1.8% | 0.39 | 0.60 | −5.9% | 0.31 |
| ShortOnly_Bear | −2.4% | −1.2% | −0.28 | −0.43 | −10.0% | −0.12 |
| **LMC (large cap)** | | | | | | |
| *Regime agnostic* | | | | | | |
| Agnostic_LongOnly | **+6.1%** | **+3.0%** | 1.11 | 1.80 | −1.9% | **1.62** |
| Agnostic_LongShort | −9.9% | −5.1% | −0.85 | −1.17 | −12.4% | −0.41 |
| Agnostic_ShortOnly | −18.5% | −9.7% | −1.62 | −2.26 | −19.4% | −0.50 |
| *Regime conditioned* | | | | | | |
| LongOnly_Bull | +4.9% | +2.4% | **1.32** | **2.10** | −2.2% | 1.08 |
| LongOnly_Bear | +3.9% | +1.9% | 0.85 | 1.39 | **−1.9%** | 1.03 |
| LongBull_ShortBear | +3.4% | +1.7% | 0.52 | 0.90 | −4.3% | 0.39 |
| ShortOnly_Bear | −1.7% | −0.9% | −0.29 | −0.49 | −6.0% | −0.14 |
| **SMC (small cap)** | | | | | | |
| *Regime agnostic* | | | | | | |
| Agnostic_LongOnly | −0.7% | −0.4% | −0.13 | −0.20 | −3.4% | −0.11 |
| Agnostic_LongShort | −2.4% | −1.2% | −0.18 | −0.27 | −10.6% | −0.11 |
| Agnostic_ShortOnly | −1.5% | −0.8% | −0.13 | −0.19 | −6.0% | −0.13 |
| *Regime conditioned* | | | | | | |
| LongOnly_Bull | **+0.9%** | **+0.4%** | **0.24** | **0.35** | **−2.2%** | **0.20** |
| LongOnly_Bear | +0.8% | +0.4% | 0.19 | 0.32 | −2.7% | 0.15 |
| LongBull_ShortBear | +0.3% | +0.2% | 0.07 | 0.09 | −6.1% | 0.03 |
| ShortOnly_Bear | −0.6% | −0.3% | −0.13 | −0.18 | −5.4% | −0.06 |

## Illustrated Trading Rules

The figures below illustrate, with made up example data, the rules that every strategy applies. Figure 12 shows all eight candlestick patterns, whether each one opens a long or a short, and the shape rule the detector checks. A candle's range is its high minus its low, its body is the distance between open and close, and its shadows are the parts of the range above and below the body.

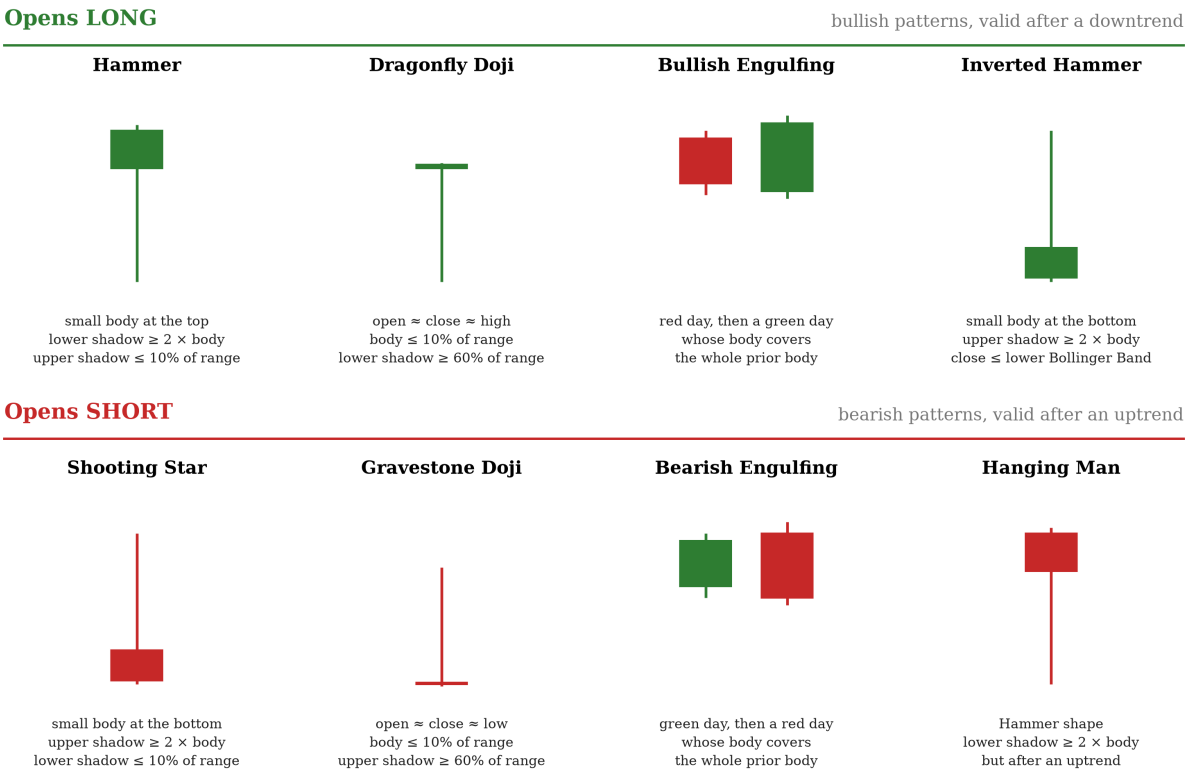



**Figure 12:** The eight candlestick patterns, grouped by the side each one opens. The top row holds the bullish patterns, which count only after a downtrend and open a long (or close a short). The bottom row holds the bearish patterns, which count only after an uptrend and open a short (or close a long). The text under each pattern gives the shape rule. Every entry must also close near the matching Bollinger Band (Figure 13) and be allowed by the strategy's regime gate.

Figure 13 shows the Bollinger Band trigger. The middle band is the moving average of the close over the Bollinger period, and the upper and lower bands sit 1.5 standard deviations above and below it. The fractional multipliers move the trigger inside the bands. A close at or below $f_L\times$ the lower band counts as near the lower band, and a close at or above $f_U\times$ the upper band counts as near the upper band. A long opens only when a bullish pattern closes near the lower band, and it can close on a bearish pattern near the upper band.

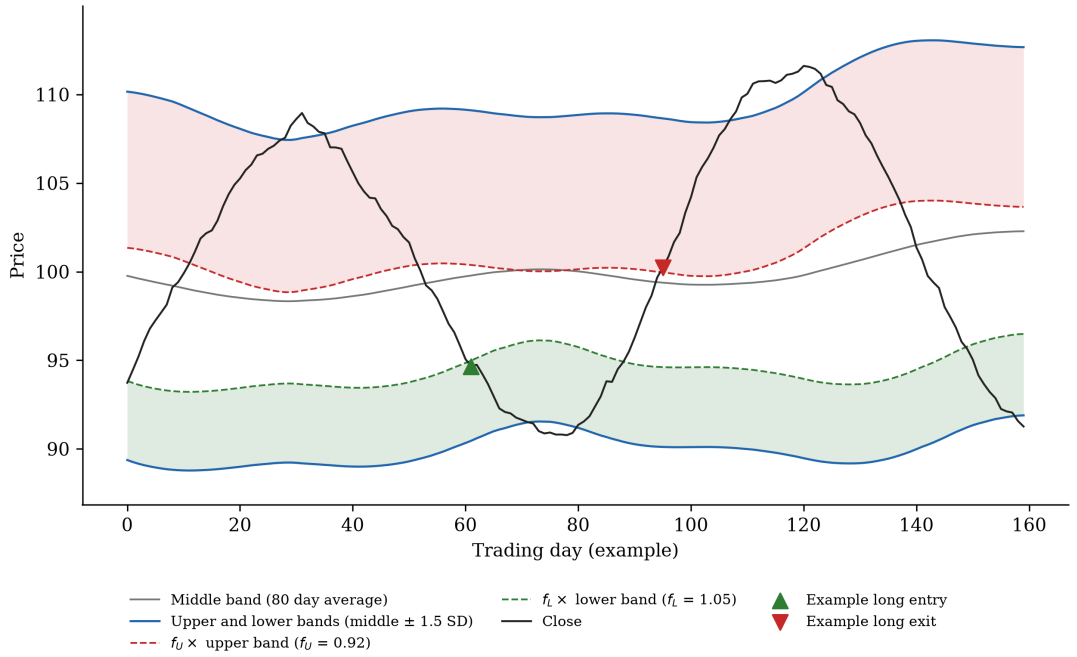



**Figure 13:** Bollinger Band trigger zones with example values of the tuned settings (80 day period, $f_L = 1.05$, $f_U = 0.92$). The green strip between the lower band and $f_L\times$ the lower band is where a close counts as near the lower band (long entry, short exit), and the red strip between $f_U\times$ the upper band and the upper band is where a close counts as near the upper band (long exit, short entry). The triangles mark an example long. In a backtest a candlestick pattern and the regime gate must also agree.

Figure 14 shows the two price based exits of a long. The profit take is fixed at entry, a set percentage above the entry price. The trailing stop follows the best close since entry and sits a set percentage below it, so it only ever moves up. Whichever level the close reaches first closes the position.

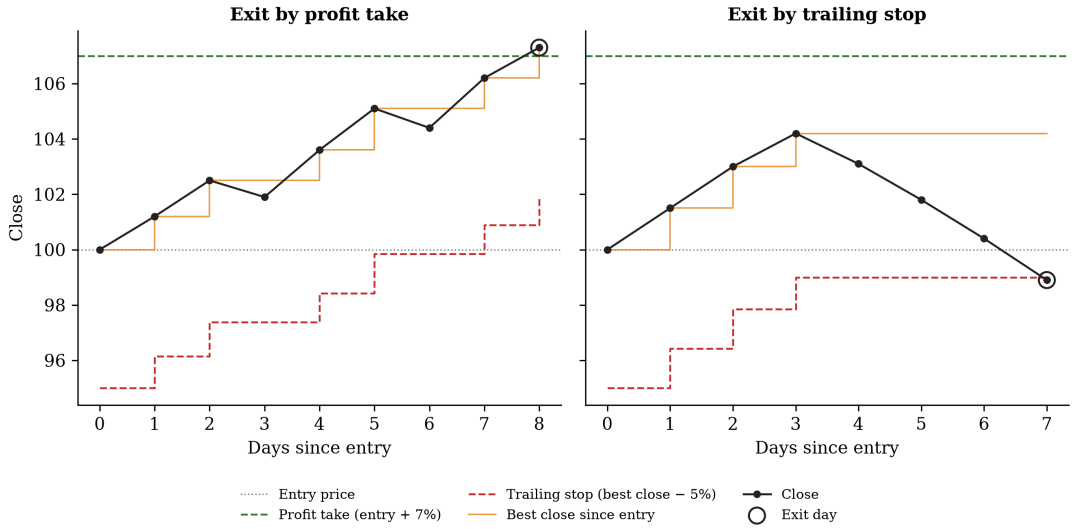



**Figure 14:** Profit take and trailing stop for a long position, with example values of 7% and 5%. Left, the close rises through the profit take level on day 8. Right, the close rises to a best close of 104.2 and then falls through the trailing stop on day 7. For a short the levels are mirrored.

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